

KESTEN'GA, Russian FederationWMO: **224030**Lat: **65.883N**Lon: **31.833E**Elev: **423**StdP: **14.47**Time Zone: **3.00 (E03)**Period: **82-90**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-16.6	-26.3	1.3	-22.1	-20.5	1.8	-16.5	19.4	27.0	17.1	26.4	1.8	330	0.695

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	75.0	62.1	71.3	60.5	68.1	58.5	63.7	72.2	61.6	69.3	59.7	66.8	6.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	79.5	67.7	58.2	73.6	64.8	56.2	68.4	62.5	29.2	72.8	27.6	69.3	26.4	67.3	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	14.0	12.3	-26.3	79.7	6.5	3.5	-31.0	82.2	-34.9	84.2	-38.5	86.2	-43.3	88.7		
			-26.5	66.8	6.5	1.7	-31.2	68.0	-34.9	69.0	-38.6	69.9	-43.3	71.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	6.3	11.0	21.4	30.3	42.5	53.0	58.3	52.7	45.3	36.0	23.7	11.2
	DBStd	19.75	15.07	12.72	9.16	7.77	7.10	8.50	5.76	6.78	6.15	7.17	10.77	13.08
	HDD50	6887	1354	1091	886	590	253	67	5	51	163	435	789	1203
	HDD65	11788	1819	1511	1351	1040	698	368	219	385	590	900	1239	1668
	CDD50	596	0	0	0	0	20	156	263	133	24	0	0	0
	CDD65	21	0	0	0	0	0	7	11	3	0	0	0	0
	CDH74	119	0	0	0	0	1	41	66	11	0	0	0	0
	CDH80	8	0	0	0	0	0	6	2	0	0	0	0	0
(14) Wind	WSAvg	5.8	5.3	5.7	5.7	5.8	5.7	5.6	5.3	5.8	5.8	7.1	6.8	5.5
(15) Precipitation	PrecAvg	22.4	1.5	1.2	1.1	1.1	1.9	2.3	3.0	2.6	2.2	1.9	1.9	1.7
	PrecMax	30.4	2.4	2.6	2.0	2.2	3.1	4.6	7.5	4.6	3.9	4.5	3.6	2.8
	PrecMin	17.2	0.3	0.3	0.6	0.3	0.6	0.9	1.8	1.0	0.4	0.6	0.2	0.5
	PrecStd	3.4	0.5	0.5	0.4	0.5	0.8	0.9	1.1	0.9	0.7	0.9	0.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.9	36.2	37.8	52.5	70.2	78.7	79.0	74.9	64.8	50.4	43.0	37.7
		MCWB	32.7	33.6	32.6	45.4	55.8	61.9	63.2	64.1	56.0	47.2	40.6	35.9
	2%	DB	32.5	33.9	35.6	47.8	63.9	74.2	75.8	70.6	59.1	48.9	40.1	34.7
		MCWB	30.8	32.2	32.7	41.5	53.2	60.9	62.7	61.5	53.7	46.3	38.6	33.5
	5%	DB	30.1	31.6	34.4	44.1	58.6	70.1	72.5	66.4	56.2	47.6	38.3	32.8
		MCWB	28.7	30.1	32.2	38.8	49.2	58.8	61.7	58.3	51.8	45.4	37.2	31.7
	10%	DB	27.1	28.9	33.1	41.4	54.5	66.4	69.1	62.9	53.9	46.1	35.9	30.4
		MCWB	26.3	27.8	31.4	36.0	46.1	57.0	59.3	56.7	50.5	44.0	34.9	29.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	34.1	34.6	46.3	57.9	64.8	66.8	65.3	57.3	49.0	41.0	36.2
		MCDB	34.5	35.4	35.7	50.6	69.1	74.2	75.5	73.4	61.9	49.6	42.0	37.1
	2%	WB	30.9	32.5	33.3	43.1	53.9	62.9	64.3	62.2	54.7	46.8	39.0	33.7
		MCDB	32.4	33.9	34.7	47.6	62.4	71.0	72.8	68.9	58.2	48.0	39.8	34.5
	5%	WB	28.9	30.2	32.5	39.0	50.2	60.2	62.0	59.8	52.5	45.7	37.2	31.7
		MCDB	30.1	31.7	34.0	43.2	57.1	69.2	70.0	65.3	55.6	47.4	38.1	32.7
	10%	WB	26.2	27.9	31.6	36.7	47.0	57.3	60.3	57.4	50.9	44.0	35.0	29.4
		MCDB	27.3	29.2	33.2	40.2	53.0	65.5	67.9	61.6	53.5	46.0	35.7	30.4
(35) Mean Daily Temperature Range		MDBR	12.1	12.2	11.9	13.5	13.5	12.9	13.1	10.5	8.8	6.9	7.9	11.7
	5% DB	MCDBR	12.8	9.9	9.5	15.4	20.8	18.0	18.6	15.0	12.8	6.6	8.1	10.8
		MCWBR	12.1	9.5	7.5	10.7	12.0	8.9	9.2	7.7	8.2	5.2	7.1	10.4
	5% WB	MCDBR	12.5	9.5	8.3	14.7	19.0	16.8	16.2	14.1	11.7	6.2	7.1	10.1
		MCWBR	11.9	9.3	6.8	10.7	11.7	9.1	9.2	8.2	8.0	5.1	6.7	9.9
(40) Clear-Sky Solar Irradiance	taub	0.204	0.270	0.301	0.333	0.347	0.343	0.365	0.342	0.317	0.290	0.353	0.202	
	taud	1.930	2.097	2.158	2.294	2.444	2.488	2.449	2.487	2.533	2.360	2.661	1.954	
	Ebn at Noon	75	185	239	263	272	276	264	260	242	185	79	11	
	Edh at Noon	6	18	28	31	30	30	30	26	20	14	6	2	
(44) All-Sky Solar Radiation	RadAvg	31	200	683	1205	1455	1635	1511	1093	601	208	45	5	
	RadStd	3	35	60	95	134	117	171	104	65	30	7	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	-277	-373	N/A	N/A

Nomenclature: See separate page